

**RESEARCH PROPOSAL**

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Building Kenyan language data infrastructure for inclusive artificial intelligence.

A proposal for a contributor-driven platform that collects, verifies, and exports Kenyan language text, translation, transcription, and speech datasets for public service, enterprise, education, health, and AI model development.

Abstract. Kenya is multilingual, yet most digital and AI systems are designed around English or broad Kiswahili coverage. This creates a language access gap for communities whose daily communication depends on indigenous Kenyan languages and regional varieties. Sema AI proposes a production-grade contributor platform for creating high-quality, consented, reviewed, and export-ready language datasets across Kenya's languages. The platform supports source corpus collection, translation, audio recording, transcription, community review, expert validation, metadata management, and dataset governance.

68+

Languages are reported in Kenya, reflecting Bantu, Nilotic, Cushitic, Semitic, and other communities.

80-90%+

Potential population benefit when local language access is added to public and private digital services.

Few

Current large public datasets cover only a small subset of Kenyan languages and domains.

1. Background and Rationale

Language is a basic access layer for technology. In Kenya, English is widely used in government, education, commerce, and formal systems, while Kiswahili serves

as a national lingua franca. However, many people continue to communicate most naturally in mother tongues, regional varieties, Sheng, and community-specific dialects. When AI tools, speech systems, chatbots, search interfaces, and public-service platforms do not support these languages, users are forced to adapt to technology instead of technology adapting to them.

The opportunity is therefore not only translation. It is the creation of a trusted national language data layer: words, phrases, sentences, domain terminology, validated translations, audio samples, transcriptions, speaker metadata, consent records, review decisions, and export pipelines that can support downstream AI systems.

2. Problem Statement

Kenyan languages remain underrepresented in digital language resources. This limits the accuracy and usefulness of machine translation, automatic speech recognition, text-to-speech, educational tools, health information systems, public-service chatbots, customer support automation, and local language AI assistants.

DATA SCARCITY

Many Kenyan languages lack large, structured, reviewed, and machine-readable corpora.

QUALITY CONTROL

Useful AI datasets need contributor identity, consent, task tracking, review, escalation, and audit trails.

SPEECH GAP

Voice systems require diverse audio across regions, genders, age groups, devices, and speaking styles.

DOMAIN COVERAGE

Health, agriculture, finance, education, law, emergency services, and government each need terminology coverage.

3. Literature Review

Recent research confirms that Kenyan NLP is advancing, but remains constrained by limited resources for indigenous languages. The 2024 survey *State of NLP in Kenya* identifies dataset creation, machine translation, speech recognition, information retrieval, and sentiment analysis as active areas, while also noting that most indigenous languages remain digitally underrepresented.

The Kencorpus project demonstrates the value of structured language data. It created text and speech resources for Swahili, Dholuo, and Luhya, including

approximately 5.6 million words and 177 hours of speech. It also produced translation and question-answer resources for downstream NLP tasks. This proves both the feasibility and importance of Kenyan language corpus development, while also showing that coverage remains limited to a small number of languages.

AfriVoices-KE extends this work into multilingual speech. The dataset includes approximately 3,000 hours of audio across Dholuo, Kikuyu, Kalenjin, Maasai, and Somali, collected from thousands of native speakers. Its methodology highlights practices that Sema AI should adopt: contributor training, regional diversity, mobile recording, scripted and spontaneous prompts, automated quality checks, and human review.

Broader research on dialect-aware NLP also shows that models trained on dominant language varieties often perform worse on dialectal and underrepresented varieties. This is directly relevant to Kenya, where a single language label may contain many regional varieties and dialects. A successful Kenyan language data platform must therefore capture language, dialect, region, speaker metadata, and task context.

4. Proposed Solution

Sema AI proposes a contributor platform and operations system for building verified Kenyan language data at scale. Contributors complete onboarding, select their strongest language, choose whether they understand English or Kiswahili source material, claim tasks, submit translations or recordings, and participate in validation workflows. Reviewers and language leads approve, reject, request changes, or escalate submissions.

- **Corpus bank:** words, phrases, sentences, idioms, proverbs, domain terms, and dialogue prompts.
- **Translation workflow:** source item, target language, dialect notes, confidence, reviewer decisions.
- **Audio workflow:** recording prompts, browser upload, metadata, playback, audio QA, and approval.
- **Review workflow:** community review, expert review, language lead decisions, audit logs.
- **Admin operations:** corpus imports, contributor oversight, recordings, translations, exports, and quality metrics.
- **Dataset export:** approved-only train/dev/test splits with metadata, consent, licensing, and provenance.

5. Beneficiaries and Expected Impact

BENEFICIARY	HOW SEMA AI HELPS
Citizens	Access health, education, civic, and business information in familiar languages.
Public institutions	Deploy multilingual digital services, helplines, and information systems.
Healthcare providers	Improve patient communication, symptom capture, triage, and translation support.
Schools and learners	Support local-language explanations, literacy, and culturally relevant learning content.
Enterprises and NGOs	Build better customer support, surveys, field operations, and outreach tools.
AI researchers and builders	Access reviewed, consented, structured Kenyan language datasets for model development.

6. Research and Implementation Questions

- Which Kenyan languages and domains should be prioritized first for maximum social and economic impact?
- How can contributor verification, consent, and quality review be designed without excluding rural users?
- What level of translation, audio, and transcription coverage is required before a language becomes useful for AI systems?
- How should dialect, region, speaker demographics, and code-switching be represented in dataset metadata?
- What governance model ensures community ownership, ethical data use, privacy, and long-term sustainability?

7. Proposed Success Metrics

- Number of active languages with verified corpus, translation, and audio coverage.
- Number of approved words, phrases, sentences, domain terms, recordings, and transcriptions.
- Reviewer agreement rate, rejection rate, revision rate, and time to approval.
- Speaker diversity across region, gender, age group, device type, and dialect.
- Dataset export readiness for machine translation, ASR, TTS, search, and assistant applications.

Core proposal: Sema AI should be developed as Kenya's contributor-driven language data and voice infrastructure platform for inclusive artificial intelligence.

8. Conclusion

The evidence suggests that Kenyan language technology is both necessary and under-resourced. Existing projects show that corpus and speech collection are feasible, but current coverage does not yet match Kenya's linguistic diversity. Sema AI can fill this gap by combining contributor participation, rigorous review workflows, audio collection, metadata governance, and export-ready datasets. The result is a platform that supports inclusion, local innovation, public-service access, and future AI systems that sound and read like Kenya.

References: Amol et al. (2024), State of NLP in Kenya: A Survey, <https://arxiv.org/abs/2410.09948> · Wanjawa et al. (2022), Kencorpus: A Kenyan Language Corpus of Swahili, Dholuo and Luhya, <https://arxiv.org/abs/2208.12081> · Wanzare et al. (2026), AfriVoices-KE: A Multilingual Speech Dataset for Kenyan Languages, <https://arxiv.org/abs/2604.08448> · Languages of Kenya, https://en.wikipedia.org/wiki/Languages_of_Kenya · Demographics of Kenya, https://en.wikipedia.org/wiki/Demographics_of_Kenya · Joshi et al. (2024), Natural Language Processing for Dialects of a Language: A Survey, <https://arxiv.org/abs/2401.05632>